

```
data <- iris

cat("Dimensions:\n")

print(dim(data))

cat("\nSummary:\n")

print(summary(data))

cat("\nStandard Deviation:\n")

print(sapply(data[1:4], sd))

cat("\nQuantiles:\n")

print(sapply(data[1:4], quantile))

cat("\nMean by Species:\n")

print(aggregate(
  . ~ Species,
  data = data,
  FUN = mean
))

cat("\nPivot Table:\n")

print(xtabs(
  ~ Species + Sepal.Length,
  data = data
))

data$Sepal_Category <- cut(
  data$Sepal.Length,
  breaks = c(0, 5, 6, 10),
  labels = c("Short", "Medium", "Long")
)

cat("\nSepal Length Categories:\n")
```

```

print(table(data$Sepal_Category))

cat("\nSpecies vs Sepal Category:\n")

print(table(

  data$Species,

  data$Sepal_Category

))

```

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for calculating mean petal width by species and creating sepal length categories.
- Environment:** Shows the 'data' object with 150 observations and 6 variables.
- Console:** Displays the output of the R code, including mean values, pivot tables, and summary statistics.

R Code in Source Editor:

```

12 ~ Species,
13 data = data,
14 FUN = mean
15 })
16 cat("\nPivot Table:\n")
17 print(xtabs(
18   ~ Species + Sepal.Length,
19   data = data
20 ))
21 data$Sepal_Category <- cut(
22   data$Sepal.Length,
23   breaks = c(0, 5, 6, 10),
24   labels = c("Short", "Medium", "Long")
25 )
26 cat("\nSepal Length Categories:\n")
27 print(table(data$Sepal_Category))
28 cat("\nSpecies vs Sepal Category:\n")
29 print(table(
30   data$Species,
31   data$Sepal_Category
32 ))
33

```

Console Output:

```

R - R 4.6.1 - ~/
3 virginica      6.588      2.974      5.552
Petal.width
1      0.246
2      1.326
3      2.026

Pivot Table:
Sepal.Length
Species 4.3 4.4 4.5 4.6 4.7 4.8 4.9 5 5.1 5.2 5.3
setosa  1  3  1  4  2  5  4  8  3  1
versicolor 0  0  0  0  0  0  1  2  1  1  0
virginica 0  0  0  0  0  0  1  0  0  0  0

Sepal.Length
Species 5.4 5.5 5.6 5.7 5.8 5.9 6 6.1 6.2 6.3 6.4
setosa  5  2  0  2  1  0  0  0  0  0  0
versicolor 1  5  5  3  2  4  4  2  3  2
virginica 0  0  1  1  3  1  2  2  6  5

Sepal.Length
Species 6.5 6.6 6.7 6.8 6.9 7 7.1 7.2 7.3 7.4 7.6
setosa  0  0  0  0  0  0  0  0  0  0  0
versicolor 1  2  3  1  1  1  0  0  0  0  0
virginica 4  0  5  2  3  0  1  3  1  1  1

Sepal.Length
Species 7.7 7.9
setosa  0  0
versicolor 0  0
virginica 4  1

Sepal Length Categories:
Short Medium Long
32 57 61

Species vs Sepal Category:
      Short Medium Long
setosa 28 22 0
versicolor 3 27 20
virginica 1 8 41
>

```